# Parabola: Short Notes

How to use these notes: read the Formulas box, follow the Steps and Key points, and avoid the Watch out points.

## 1. Conic: definition and general equation

**Formulas**

$$PS = e \cdot PM$$

$$\Delta = abc + 2fgh - af^2 - bg^2 - ch^2$$

KEY POINTS

- Focus $S$, directrix, eccentricity $e$; axis $\perp$ directrix through $S$; vertex on axis; $Z$ = foot of directrix
- $\Delta = 0$: pair of lines; $\Delta \neq 0$: $h^2 = ab$ parabola, $h^2 < ab$ ellipse, $h^2 > ab$ hyperbola
- Focus on directrix gives lines; otherwise $e = 1, < 1, > 1$: parabola, ellipse, hyperbola
- $PS = e\,PM$ form: divide by $\sqrt{l^2 + m^2}$ to read $S$, directrix and $e$

**Watch out**

$h^2 = ab$ alone is not enough: check $\Delta \neq 0$

## 2. Standard parabola $y^2 = 4ax$

**Formulas**

$S(a, 0)$, directrix $x = -a$, LR $= 4a$

$$PS = x_1 + a$$

KEY POINTS

- Vertex $O$, axis $y = 0$, tangent at vertex $x = 0$, $Z(-a, 0)$, LR $x = a$, ends $(a, \pm 2a)$
- LR $= 2\times$ (focus to directrix); vertex = mid-point of $SZ$
- Focal chord: through $S$; double ordinate: $\perp$ axis; LR: double ordinate through $S$
- Circle touching a fixed circle and a fixed line: centre on a parabola (focus = centre of fixed circle, directrix = line shifted by $r$)

## 3. Other standard and shifted parabolas

**Formulas**

$(y - k)^2 = 4a(x - h)$, $(x - h)^2 = 4a(y - k)$

KEY POINTS

- $y^2 = -4ax$: $S(-a, 0)$, $x = a$; $x^2 = 4ay$: $S(0, a)$, $y = -a$; $x^2 = -4ay$: $S(0, -a)$, $y = a$
- One square term, no $xy$: shifted parabola; complete the square
- $(y - k)^2 = 4a(x - h)$: focus $(h + a, k)$, directrix $x = h - a$, LR ends $(h + a, k \pm 2a)$

STEPS

1. Put $X = x - h$, $Y = y - k$
2. Read results for $Y^2 = 4aX$
3. Shift back to $x, y$

**Watch out**

$a < 0$ reverses the opening; LR $= 4|a|$

## 4. Oblique axis: focus-directrix and alternate definition

**Formulas**

$$\left(\frac{L_1}{\sqrt{l^2 + m^2}}\right)^2 = 4a \cdot \frac{L_2}{\sqrt{l^2 + m^2}}$$

KEY POINTS

- $L_1 = 0$ axis, $L_2 = 0$ tangent at vertex ($L_1 \perp L_2$); (distance from axis)$^2$ = LR $\times$ (distance from tangent at vertex)

- Focus + directrix given: write $PS^2 = PM^2$ directly
- Vertex + directrix: $a$ = distance of vertex from directrix; focus = reflection of $Z$ in the vertex

**Watch out**

Choose the sign of $L_2$ so that the focus side is positive

## 5. Parametric point and focal chord

**Formulas**

$$P(at^2, 2at)$$

Focal chord: $t_1 t_2 = -1$, length $a\left(t + \frac{1}{t}\right)^2 = 4a\,\mathrm{cosec}^2\alpha$

KEY POINTS

- $m_{OP} = \frac{2}{t}$; chord slope $\frac{2}{t_1+t_2}$; double ordinate $t, -t$; LR ends $t = \pm 1$
- Right angle at vertex: $t_1 t_2 = -4$; chord through $(c, 0)$: $t_1 t_2 = -\frac{c}{a}$
- Focal segments $a(1+t^2)$, $a(1+\frac{1}{t^2})$; $\frac{1}{SP} + \frac{1}{SQ} = \frac{1}{a}$
- Distance of focal chord from vertex $= a\sin\alpha$; least focal chord = LR

STEPS

1. Locus: take $(at^2, 2at)$
2. Write $h, k$ in $t$
3. Eliminate $t$

## 6. Position of point, chords

**Formulas**

$$S_1 = y_1^2 - 4ax_1$$

Chord: $2x - (t_1 + t_2)y + 2at_1t_2 = 0$

$$\frac{4}{m^2}\sqrt{a(1+m^2)(a-mc)}$$

KEY POINTS

- $S_1 < 0$ inside (focus side), $= 0$ on, $> 0$ outside
- Chords subtending $90°$ at vertex pass through $(4a, 0)$
- Angle at vertex: homogenise $y^2 = 4ax$ with the line; $\tan\theta = \frac{2\sqrt{H^2 - AB}}{|A+B|}$

## 7. Tangent

**Formulas**

$$yy_1 = 2a(x + x_1)$$

$$ty = x + at^2$$

$y = mx + \dfrac{a}{m}$, contact $\left(\dfrac{a}{m^2}, \dfrac{2a}{m}\right)$

KEY POINTS

- $m = \frac{1}{t}$; tangents at $t_1, t_2$ meet at $(at_1t_2,\ a(t_1+t_2))$
- From $(x_1, y_1)$: $x_1m^2 - y_1m + a = 0$; $m_1 + m_2 = \frac{y_1}{x_1}$, $m_1m_2 = \frac{a}{x_1}$
- Perpendicular tangents meet on the directrix (director circle); ends of a focal chord give perpendicular tangents
- Line vs parabola: $m^2x^2 + 2(mc - 2a)x + c^2 = 0$, $D > 0$ secant, $D = 0$ tangent, $D < 0$ none

**Watch out**

Line parallel to the axis cuts once but is not a tangent

## 8. Normal

**Formulas**

$$y = -tx + 2at + at^3$$

$y = mx - 2am - am^3$, foot $(am^2, -2am)$

$$t_2 = -t_1 - \frac{2}{t_1}$$

KEY POINTS

- At a point: $y - y_1 = -\frac{y_1}{2a}(x - x_1)$; $m = -t$
- $y = mx + c$ normal $\iff$ $c = -2am - am^3$
- Normals at $t_1, t_2$ meet at $\big(2a + a(t_1^2 + t_1t_2 + t_2^2),\ -at_1t_2(t_1 + t_2)\big)$
- Normals at $t_1, t_2$ meet on the curve: $t_1t_2 = 2$
- Shortest distance point-curve: along a normal; line-curve: parallel tangent

## 9. Normals from a point (co-normal points)

**Formulas**

$$am^3 + (2a - h)m + k = 0$$

KEY POINTS

- At most 3 normals; $\sum m = 0$, $\sum m_1m_2 = \frac{2a-h}{a}$, $m_1m_2m_3 = -\frac{k}{a}$
- Feet $t_1 + t_2 + t_3 = 0$; centroid on the axis
- Circle through the feet: $2(x^2 + y^2) - 2(h + 2a)x - ky = 0$ (passes through vertex)
- Three real normals from $(h, 0)$ need $h > 2a$

STEPS

1. Locus with a condition on slopes: write Vieta relations
2. Find the remaining root and put it in the cubic
3. Replace $(h, k)$ by $(x, y)$

## 10. Transformation rules

KEY POINTS

- $y^2 = -4ax$: $a \to -a$; $x^2 = 4ay$: swap $x, y$; shifted: $x \to x - h$, $y \to y - k$
- Tangency $y = mx + c$: $y^2 = -4ax$: $c = -\frac{a}{m}$; $x^2 = 4ay$: $c = -am^2$; $x^2 = -4ay$: $c = am^2$
- Normal $y = mx + c$: $y^2 = -4ax$: $c = 2am + am^3$; $x^2 = 4ay$: $c = 2a + \frac{a}{m^2}$; $x^2 = -4ay$: $c = -2a - \frac{a}{m^2}$
- Shifted: tangent $y - k = m(x - h) + \frac{a}{m}$, normal $y - k = m(x - h) - 2am - am^3$

**Watch out**

Slope changes when $x, y$ are swapped: re-derive, do not copy $c = \frac{a}{m}$

## 11. Common tangents and normals

STEPS

1. Write the line as $y = mx + \frac{a}{m}$ for the parabola
2. Apply the other curve's condition (circle: $p = r$; parabola/ellipse: its $c$-condition)
3. Solve for $m$; points of contact last

KEY POINTS

- Use symmetry (e.g. curves symmetric about $y = x$ meet their common tangent of slope $\pm 1$)
- Common normal of two curves = shortest distance between them

## 12. Chord of contact, mid-point chord, pair of tangents

**Formulas**

$T = 0$; $T = S_1$; $SS_1 = T^2$

$L = \dfrac{\sqrt{S_1(y_1^2 + 4a^2)}}{a}$, Area $= \dfrac{S_1^{3/2}}{2a}$

KEY POINTS

- Mid-point $(h, k)$ of a chord of slope $m$: $k = \frac{2a}{m}$
- Chord of contact $lx + my + n = 0$ comes from $\left(\frac{n}{l}, -\frac{2am}{l}\right)$
- Points on a fixed line: chords of contact pass through its pole

## 13. Pole, polar, diameter

**Formulas**

Polar: $yy_1 = 2a(x + x_1)$; pole of $lx + my + n = 0$: $\left(\dfrac{n}{l}, -\dfrac{2am}{l}\right)$

Diameter: $y = \dfrac{2a}{m}$

KEY POINTS

- $T = 0$: tangent (on), chord of contact (outside), polar (always)
- Polar of focus = directrix; every diameter is parallel to the axis

## 14. Highlights (properties)

KEY POINTS

- Tangent between contact and directrix subtends $90°$ at $S$
- Foot of $\perp$ from $S$ on a tangent lies on the tangent at vertex; image of $S$ in a tangent lies on the directrix
- Tangents at ends of a focal chord: perpendicular, meet on directrix
- Circle on focal chord touches directrix; circle on focal distance touches tangent at vertex
- $T, G$ on axis: $ST = SG = SP$; $S$ is circumcentre of $TPG$; reflection property
- Triangle area $a^2|(t_1 - t_2)(t_2 - t_3)(t_3 - t_1)| = 2\times$ triangle of tangents
- LR ends: tangents meet at $(-a, 0)$, normals at $(3a, 0)$, forming a square

## 15. Common mistakes checklist

STEPS

1. Shifted parabola: shift the focus/directrix back; LR uses $|a|$
2. $x^2 = 4ay$ results: swap $x, y$ before using $c = \frac{a}{m}$
3. Tangent from a point: check the point is outside ($S_1 > 0$)
4. Normal from a point: the cubic in $m$ may have only one real root
5. Focal chord: $t_1 t_2 = -1$; right angle at vertex: $t_1 t_2 = -4$ (do not mix)
6. Shortest distance: discard feet that give a maximum or complex $t$